# Problem Set 1

Preferences and Utility Functions

EC306: Intermediate Microeconomics

**Instructions.** Work through these in class. Show all working and interpret your numerical answers in words. Worked solutions follow at the end.

## Question 1 — Axioms and their limits

(a) State the completeness and transitivity axioms formally.

(b) A consumer ranks three holidays: Lisbon $\succ$ Athens, Athens $\succ$ Rome, Rome $\succ$ Lisbon. Construct an explicit "money-pump" sequence of three trades, each improving the consumer's bundle by their own ranking, that returns them to the start strictly poorer. State the total loss if each trade costs £$c$.

(c) In lecture we argued the axioms are a modelling device, not literal truths. Give *one* concrete, named real-world phenomenon (e.g. a documented behavioural effect) that violates transitivity, and explain in 3–4 sentences why economists nonetheless retain the axiom. Generic answers ("people are irrational") score zero; name the effect and tie it to the axiom.

## Question 2 — MRS, convexity, and a general CES

Consider the CES utility $u(x, y) = (\alpha x^\rho + (1-\alpha)y^\rho)^{1/\rho}$, with $0 < \alpha < 1$ and $\rho < 1,\ \rho \neq 0$.

(a) Derive $\text{MU}_x$, $\text{MU}_y$, and show that $\text{MRS} = \dfrac{\alpha}{1-\alpha}\left(\dfrac{x}{y}\right)^{\rho-1}$.

(b) Show that as $\rho \to 1$ the MRS becomes independent of $(x, y)$, and interpret what kind of goods this limiting case describes.

(c) Take the limit $\rho \to 0$. Show the preferences converge to Cobb–Douglas and identify the resulting exponents. (Hint: examine $\ln u$ and apply l'Hôpital in $\rho$.)

(d) For $\rho = -1$, $\alpha = 0.5$, evaluate the MRS at $(x, y) = (1, 4)$ and at $(4, 1)$. In two sentences, explain what the *change* in the MRS between these points tells you about how substitutable the two goods are here, relative to the Cobb–Douglas case.

## Question 3 — Representation and transformations

(a) Define precisely what it means for $u(\cdot)$ to *represent* a preference relation.

(b) You are given two utility functions:

$$u_1(x, y) = x^{1/3}y^{2/3}, \qquad u_2(x, y) = \ln x + 2\ln y.$$

Determine whether they represent the same preferences. If so, exhibit the exact monotonic transformation $g$ such that $u_2 = g(u_1)$; if not, prove they differ by finding a pair of bundles ranked oppositely.

(c) Now consider $u_3(x, y) = x^{1/3}y^{2/3} + 0.01x$. Is $u_3$ a monotonic transformation of $u_1$? Justify carefully — compute and compare the two MRS expressions, and explain what economically changes when the extra term is added.

## Question 4 — Functional forms meet the real market

This question requires **current data you locate yourself**. Cite each source with a URL and the date you accessed it.

(a) Find the *current* retail price of one specific inkjet printer model and a single compatible OEM ink cartridge for it. Report both prices, the model, and your sources.

(b) Compute the ratio (annualised cartridge spend for a stated, justified usage level) to (printer price). State your usage assumption explicitly.

(c) Using the perfect-complements model from lecture, explain why the observed pricing pattern (cheap hardware, expensive consumable) is consistent with $u = \min\{x, y\}$-type demand. Be specific about which good carries the margin and why the fixed-ratio property enables this.

(d) Identify *one* feature of the real market that the pure perfect-complements model misses (e.g. third-party cartridges, refilling, chipped cartridges). Explain in 3–4 sentences how that feature would change the consumer's effective MRS and therefore the firm's ability to extract margin.